MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **WEEKLY CHAPTER TEST – 06 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If one zero of the polynomial $(a^2+9)x^2+13x+6a$ is reciprocal of the other, then the value of a is: [1]

(A) 3 (B) -3 (C) $\frac{1}{3}$ (D) 9

Q2. If the zeroes of the polynomial x^2-bx+c are two consecutive positive integers, then b^2-4c is: [1]

(A) 1 (B) 2 (C) -1 (D) 0

Q3. If one zero of $p(x)=ax^2+bx+c$ is double the other, then the relation between a, b, c is: [1]

(A) $2b^2=9ac$ (B) $9b^2=2ac$ (C) $b^2=4ac$ (D) $2a^2=9bc$

Q4. The value of k for which the zeroes of the polynomial $p(x)=9x^2-12x+k$ are equal, is: [1]

(A) 4 (B) -4 (C) 3 (D) 2

Q5. **Assertion (A):** If α, β are zeroes of kx^2+4x+4 such that $(\alpha+\beta)^2-2\alpha\beta=24$, then $k=1$ is an integer solution.
Reason (R): For kx^2+4x+4, the sum of zeroes is $-\frac{4}{k}$ and the product of zeroes is $\frac{4}{k}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type Questions (1 × 2 + 3 × 3 = 11 Marks)

Q6. Find the condition on the coefficients of ax^2+bx+c such that its zeroes are equal in magnitude but opposite in sign. [2]

Q7. If α, β are the zeroes of the polynomial $3x^2-2x-1$, obtain a quadratic polynomial whose zeroes are $\alpha+2\beta$ and $2\alpha+\beta$. [3]

Q8. If α and β are the zeroes of the quadratic polynomial $2x^2+3x+1$, obtain a quadratic polynomial whose zeroes are $\alpha-2\beta$ and $\beta-2\alpha$. [3]

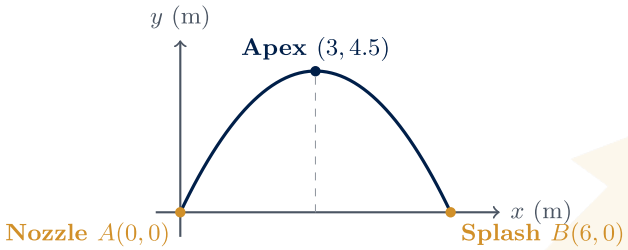
Q9. Find the zeroes of the polynomial $v(x)=v^2+4\sqrt{3}v-15$ by factorisation and verify the relationship between zeroes and coefficients. [3]

SECTION C: Applied Case Study Analysis (1 × 4 = 4 Marks)

Question 10: Case Study Analysis — Botanical Garden Water Fountain

In a decorative botanical garden, a water fountain shoots a stream of water from nozzle A located at the edge of a pond ($x = 0$). The water rises, attains a peak height, and falls into the pond at point B. The height y (in metres) of the water stream at horizontal distance x (in metres) from the nozzle is given by:

$$y = -\frac{1}{2}x^2 + 3x$$



Based on the above hydraulic model, answer the following questions:

(i) Find the horizontal distance between the nozzle A and the splash point B. [1]

(ii) What is the maximum height attained by the water stream above the pond level? [1]

(iii) **(a)** At what horizontal distance(s) from nozzle A is the water stream at a height of 4 m? [2]

— **OR** —

(b) If the nozzle is adjusted so that the new water jet is modelled by $y = -x^2 + 4x$, calculate its new maximum height and horizontal range. [2]

MATHS MANTHAN TUITION CLASSES

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★